561

Visual Design Bible

A medium-independent specification for typography, color, information structure, institutional habitats, LaTeX/PDF publication, TUFTE projection, and reproducible design computation.

VERSION	0.4.0-candidate
STATUS	Candidate specification; prepared from recovered 561 implementation evidence and current design directives.
BUILD PROFILE	XeLaTeX / R-derived tokens / A4 / two-sided / artifact-ready
CORE PRINCIPLE	2047, but comfortable: futuristic through capability and information, not decorative sci-fi noise.

Document control

SPECIFICATION	CANDIDATE
TOKEN VALIDATION	PASS
RECOVERED CORE SET	8 CORE COLORS

NORMATIVE

This document is both a design manual and an executable design specimen. The rules that govern this PDF are the same rules being specified. The intended canonical source is not the prose alone: it is the combination of (1) medium-independent design tokens, (2) normative semantic rules, (3) renderer profiles such as `561design.sty`, and (4) conformance tests.

The 561 Group already has a coherent visual language, but it exists in several layers of evidence rather than one accepted specification. The current web-estate registry explicitly warns that historical repositories, deployment state, and self-declarations are evidence rather than authority. This Bible therefore treats recovered implementations as observations, reconciles them with current Group direction, and labels new decisions as candidate rules rather than pretending that every old CSS declaration was canonical.

What this document is for

The Bible exists so that a person, agent, renderer, or build system can answer questions such as:

- Which visual decisions are shared across the Group, and which belong to a habitat such as Lenticule Science or Coesite Operations?
- Which values are machine-readable tokens rather than ad-hoc styling choices?
- How should the same semantics render into web, LaTeX/PDF, SVG, print, and future TUFTE projections?
- What does a conforming 561 document look like when no designer is present to “make it look right” by hand?
- Which choices are prohibited because they create drift, reduce accessibility, or substitute fashionable appearance for durable identity?

Normative vocabulary

The verbs **MUST**, **MUST NOT**, **SHOULD**, **SHOULD NOT**, and **MAY** are used as conformance language. This is intentionally stricter than a mood board. A design system that cannot reject a bad build is only a collection of preferences.

Authority model

Layer	Status in this Bible	Treatment
Current Group directives	normative input	Preserved unless internally inconsistent or impossible to render accessibly.
Web-estate registry	strong governance evidence	Defines web-estate control-root and TUFTE direction; explicitly rejects accidental authority by deployment or README.
Current 561 theme package	recovered implementation evidence	Strong evidence for palette, type families, hard hierarchy, folio/hero/colophon, and zero-radius geometry.
Current small static apex	deployment evidence	Useful for reading measure, gutters, dark-mode values, and minimal content styling; not canonical identity by itself.
Historical Hugo theme	provenance	Strong evidence for Bauhaus/Brutalist/Tuftean intent, evidence typography, document surfaces, bibliography, print behavior, and reduced-motion handling.
This Bible v0.4	candidate synthesis	Proposed single source of visual truth pending institutional adoption and implementation feedback.

SOURCE: https://github.com/561-group/561-group/blob/main/sites/README.md

SOURCE: https://github.com/561-group/561-group/blob/main/sites/registry.json

SOURCE: https://github.com/561-group/561-group-website-theme/blob/main/src/index.mts

SOURCE: https://github.com/561-group/561-group-site/blob/main/public/index.html

SOURCE: https://github.com/emsenn/561-hugo-theme/blob/main/static/style.css

Contents

Document control i

The twelve rules viii

1 Precedent: what a real design Bible is 1

1.1 Reference model . . . 2

1.2 The 561 document class . . . 3

2 Recovered 561 visual strata 4

2.1 The strongest inherited signals . . . 4

2.2 Recovered values that become candidate core . . . 5

3 North star: 2047, but comfortable 6

3.1 Positive definition . . . 6

3.2 Negative definition . . . 7

3.3 Three simultaneous readings . . . 7

4 Identity architecture 8

4.1 One house, multiple habitats . . . 8

4.2 Inheritance law . . . 8

4.3 Marks and identifiers . . . 8

5 Token architecture 10

5.1 Why tokens are the source of truth . . . 10

5.2 Four token tiers . . . 10

5.3 Naming rules . . . 10

5.4 Projection contract . . . 11

6 Color system 12

6.1 Core colors . . . 12

6.2 Semantic roles . . . 12

6.3 Computed contrast . . . 13
6.4 Dark mode . . . 13

7 Typography 14
7.1 Three channels . . . 14
7.2 Specimen . . . 14
7.3 Scale . . . 15
7.4 Line length and rhythm . . . 15

8 Spatial system and grid 16
8.1 Spacing ladder . . . 16
8.2 Grid law . . . 16
8.3 A4 print profile . . . 16
8.4 Web profile . . . 17

9 Rules, edges, shape, and texture 18
9.1 Rules are structural language . . . 18
9.2 Corners . . . 18
9.3 Texture . . . 18

10 Information design 19
10.1 Tuftean, not Tufte cosplay . . . 19
10.2 Captions are evidence . . . 19
10.3 Tables . . . 19
10.4 Figures and diagrams . . . 19
10.5 Data visualization . . . 20

11 Imagery and visual assets 21
11.1 Asset order of operations . . . 21
11.2 Photography . . . 21
11.3 Generated imagery . . . 21
11.4 Deterministic visual macros . . . 22

12 Interaction and motion 23
12.1 Motion has a job . . . 23
12.2 Focus is visible architecture . . . 23
12.3 Links . . . 23

13 Accessibility and legibility **24**

13.1 Accessibility is upstream . . . 24

13.2 Color . . . 24

13.3 Type and measure . . . 24

13.4 PDF accessibility . . . 24

14 Component grammar **25**

14.1 Four primary containers . . . 25

14.2 Cards are not a default . . . 25

14.3 Forms . . . 25

14.4 Status and metadata . . . 26

15 Math, code, proof, and technical documents **27**

15.1 Technical material is first-class . . . 27

15.2 Mathematics . . . 27

15.3 Code . . . 27

15.4 Proof/status apparatus . . . 28

16 Habitat profile: 561 Group apex **29**

16.1 Material cues . . . 29

16.2 Web chrome . . . 29

17 Habitat profile: Lenticule Science **30**

17.1 Expression . . . 30

18 Habitat profile: Red Cup Engineering **31**

18.1 Expression . . . 31

19 Habitat profile: Coesite Operations **32**

19.1 Expression . . . 32

20 Habitat profile: Causeway Public Services **33**

20.1 Expression . . . 33

21 Habitat profile: Koios.News **35**

22 Habitat profile: Turntide Islands **36**

22.1 Provisional expression . . . 36

23 LaTeX as a first-class renderer **37**

23.1 Why LaTeX . . . 37

23.2 Repository layout . . . 37

23.3 Package responsibilities . . . 37

23.4 Renderer API . . . 37

24 R as design computation **39**

24.1 Why R belongs in the build . . . 39

24.2 Current build computation . . . 39

24.3 Next computations to add . . . 39

25 GitHub Actions: source to PDF artifact **41**

25.1 Build contract . . . 41

25.2 CI acceptance . . . 41

26 Conformance profiles **42**

26.1 Profile matrix . . . 42

26.2 Claiming conformance . . . 42

26.3 Golden specimens . . . 42

27 Governance and versioning **44**

27.1 Design changes are engineering changes . . . 44

27.2 Decision classes . . . 44

27.3 No silent drift . . . 44

27.4 Visual debt . . . 45

28 Migration plan from what exists now **46**

28.1 Phase 1: admit the Bible repository . . . 46

28.2 Phase 2: unify tokens . . . 46

28.3 Phase 3: cross-medium proof . . . 46

28.4 Phase 4: habitat packages . . . 47

28.5 Phase 5: conformance service . . . 47

29 Implementation checklist **48**

29.1 For a new LaTeX document . . . 48

29.2 For a new visual component . . . 48

29.3 For a new habitat . . . 48

30 Token registry **50**

31 Build and release recipe **51**

31.1 Local build 51

31.2 Expected products 51

31.3 Preflight 51

32 References and precedents **52**

Colophon **54**

The twelve rules

ONE-PAGE OPERATING DOCTRINE

If a renderer remembers only one spread, remember this one.

1. **Semantics precede styling.** Content/state becomes a form model, then topology, then medium-specific styling. CSS, LaTeX, and PDF are projections, not identity authorities.
2. **The house is 2047, but comfortable.** It may feel wildly advanced, but never hostile, generic-cyberpunk, glossy-startup, or synthetic for its own sake.
3. **Type is mass.** Display sans carries institutional force; slab serif carries reading; mono carries evidence, apparatus, labels, code, and receipts.
4. **Bone and ink dominate.** Witness blue and alarm red are scarce semantic devices, not decorative paint.
5. **One region, one salience.** Do not make every element loud. The eye should know what matters before it reads.
6. **Hard hierarchy beats ornament.** Use scale, spacing, rules, alignment, and typographic contrast before adding decoration.
7. **Every habitat is related, not identical.** 561, Lenticule, Red Cup, Coesite, Causeway, Koios, and other properties inherit shared grammar but may express mission-specific material worlds.
8. **Figures are evidence.** Captions say what a figure demonstrates, its source/provenance, and limits. Decorative illustration never substitutes for information.
9. **Accessibility is a design input.** Contrast, focus, motion, reading measure, hierarchy, print behavior, alt text, and semantic structure are conformance requirements.
10. **The system is tokenized.** Core decisions live in DTCG-compatible design tokens and are transformed into medium profiles.

11. **Builds are reproducible.** R computes/validates design data; LaTeX renders; CI checks; artifacts are versioned. No human hand-tuning after the build is the normal path.
12. **No parallel branding systems.** New needs extend the shared system or explicitly declare a habitat exception; they do not silently invent another identity.

SECTION 1

Precedent: what a real design Bible is

The closest real-world analog is not an ordinary "brand book." A durable design Bible combines the authority and exactness of a graphics standards manual, the reusable components and tokens of a design system, the accessibility obligations of a public service standard, and the build/test discipline of software infrastructure.

1.1 Reference model

Precedent	What 561 should borrow	What not to copy blindly
NASA Graphic Standards	A single comprehensive identity authority; exact identifier construction/placement; consistency across media.	NASA's marks, exact aesthetic, or institution-specific legal restrictions.
USWDS	Design principles, tokens, accessibility as foundation, continuity across services, user needs and trust as system-level constraints.	Federal-government visual voice as such.
GOV.UK Design System	Explicit spacing/-type scales, reuse-before-invention, conventions with strong implementation guidance.	GOV.UK's specific brand appearance.
DTCG 2025.10	Vendor-neutral JSON token source; aliases/groups; platform-agnostic design decisions.	Treating tokens as the whole design system; they are only one layer.
WCAG 2.2	Contrast, focus, motion and interaction constraints as acceptance criteria.	Reducing accessibility to contrast alone.
Carbon 2x Grid	A geometric foundation tying layout, typography, iconography, and illustration together.	IBM's product visual identity.
ISO 216 + PDF/A	Stable page geometry and archival-output thinking.	Assuming one trim or archival profile suits every publication.

EVIDENCE / RATIONALE

NASA describes its graphics standards as a means of maintaining a clear, consistent visual identity. USWDS explicitly treats accessibility as fundamental and tokens as discrete palettes for typography, spacing, color, and other style decisions. GOV.UK publishes explicit spacing and type scales. The Design Tokens Community Group published its first stable interoperable format in October 2025. These are the behaviors this Bible is trying to emulate: visible rules, machine reuse, conformance, and maintenance.

SOURCE: https://www.nasa.gov/nasa-brand-center/brand-guidelines/

SOURCE: https://designsystem.digital.gov/design-principles/

SOURCE: https://designsystem.digital.gov/design-tokens/

SOURCE: https://design-system.service.gov.uk/styles/

SOURCE: https://www.w3.org/community/reports/design-tokens/CG-FINAL-format-20251028/

SOURCE: https://www.w3.org/WAI/standards-guidelines/wcag/

SOURCE: https://www.carbondesignsystem.com/building-blocks/foundations/2x-grid/guidelines

SOURCE: https://www.iso.org/standard/36631.html

SOURCE: https://pdfa.org/resource/iso-19005-pdfa/

1.2 The 561 document class

A mature 561 visual specification should have four coupled forms:

1. **Bible**: human-readable rationale, examples, prohibitions, and visual plates.
2. **Tokens**: machine-readable decisions with versioning and aliases.
3. **Renderer profiles**: LaTeX package, CSS/TUFTE theme, SVG/TikZ macros, presentation profile.
4. **Conformance suite**: automated checks plus a small set of golden visual specimens.

NORMATIVE

A visual rule is not fully admitted until it has a human explanation, a token or renderer representation where applicable, and at least one conformance method. Rules that cannot yet meet that bar remain informative or experimental.

SECTION 2

Recovered 561 visual strata

2.1 The strongest inherited signals

The recovered material is remarkably consistent despite being distributed across generations:

- Bauhaus composition, brutalist typography, constructivist poster logic, Cubist fragmentation where useful, and Tuftean information design.
- Few colors; scarce salience; bone/paper ground; ink marks; one warm alarm red; one witness/reference blue.
- Wide reading measure discipline, strong rules, square geometry, and clear folio/metadata furniture.
- Typography split by role: expressive/institutional display, long-form reading, and monospaced evidence/apparatus.
- Print is not an afterthought: historical CSS already removes motion, flattens document cards, exposes URLs, and protects figures from bad page breaks.
- The current web control-root says the final 561.group projection should derive from structured source; visual style is a parameter of the projection, not a standalone static theme.

2.2 Recovered values that become candidate core

Decision	Recovered evidence	Candidate decision
Core ink	#0A0A0A in current theme	Adopt as color.ink.
Core ground	#F2EAD8 in current theme	Adopt as color.bone.
Witness blue	#3D5588 current theme	Adopt as general 561 evidence/reference blue.
Alarm red	#FF4530 current theme	Adopt with stricter accessibility/use constraints.
Display family	Space Grotesk named in theme source	Adopt as preferred; renderer must provide local fallback.
Reading family	Roboto Slab named in theme source	Adopt as preferred core reading face.
Evidence family	JetBrains Mono named in theme source	Adopt as preferred evidence/code face.
Web max/gutter	1200px / 56px current theme	Adopt as web defaults, not universal geometry.
Reading measure	42rem current site/theme	Adopt as web long-form target.
Corner radius	0 current theme	Adopt as core; exceptions must belong to a habitat/component need.

EXCEPTION / CONSEQUENCE

The recovered 561 website-theme repository is itself listed as quarantine in the web-estate registry. This Bible does not silently promote that package to authority. It promotes specific values only where they agree with current direction and form a coherent cross-medium system.

SECTION 3

North star: 2047, but comfortable

HOUSE CHARACTER

Wildly futuristic; humanly inhabitable.

The 561 house should look like an institution that has spent twenty years learning how to make complex systems legible, rather than a company that discovered a fashionable science-fiction skin last quarter. The future is conveyed through capability: better information structure, live evidence, unusual but coherent instruments, rigorous diagrams, computational typography, and calm interaction with sophisticated systems.

3.1 Positive definition

NORMATIVE

A conforming 561 surface **MUST** feel durable, information-dense, purposeful, tactile, and habitable. It **SHOULD** combine modernist institutional clarity with enough strangeness to imply that the system is doing work ordinary interfaces do not yet do.

The material vocabulary is allowed to evoke paper, enamel, limestone, concrete, brushed metal, lab glass, archival folders, engineering drawings, phosphor/instrument displays, transit signage, civic forms, field notebooks, and scientific plates. Digital surfaces may be highly dynamic, but their underlying geometry should remain comprehensible.

3.2 Negative definition

EXCEPTION / CONSEQUENCE

The default 561 style **MUST NOT** collapse into generic “AI startup” presentation: no ambient purple/blue gradient fog, glassmorphism as filler, feature-card monoculture, random glowing orbs, faux-terminal decoration, undifferentiated dashboard grids, or cyberpunk noise used merely to signal futurity.

Nor should “brutalist” mean careless. Hard edges, large type, visible rules, and dense apparatus are only valuable when the hierarchy is more legible as a result.

3.3 Three simultaneous readings

A strong 561 page should work at three distances:

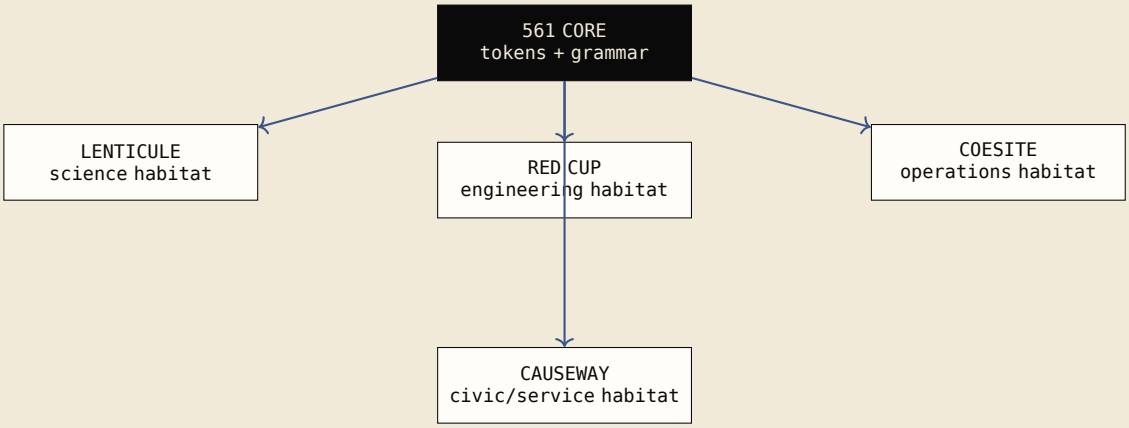
1. **Across the room**: silhouette, dominant hierarchy, identity, and main state are obvious.
2. **At reading distance**: argument and navigation are calm, generous, and legible.
3. **Under inspection**: provenance, metadata, data, citations, state, and machine-readable structure are available without a second visual language.

SECTION 4

Identity architecture

4.1 One house, multiple habitats

The Group is not a monobrand painted onto every subsidiary. It is a shared grammar that supports distinct institutional habitats. Continuity comes from type channels, evidence furniture, token naming, rules, layout logic, accessibility, and provenance—not from forcing every site to use exactly the same background and accent.



4.2 Inheritance law

NORMATIVE

A habitat **MAY** override surface color, material cues, display composition, iconography, and certain component details. It **MUST** inherit semantic token roles, typography channels or approved equivalents, accessibility thresholds, provenance treatment, content-state grammar, and the no-parallel-branding rule.

4.3 Marks and identifiers

The giant “561” mark in recovered web chrome is useful as a typographic institutional device. It is not sufficient to define a corporate identifier standard. A later mark annex should specify minimum size, exclusion zone, small-format behavior,

lockups, monochrome use, reproduction tests, and whether the three-digit mark is formally a logo or merely display typography.

Until that annex exists:

- Use the wordmark/name and large “561” typographic mark conservatively.
- Do not invent alternate emblems, mascots, badges, seals, or subbrand logos merely to fill space.
- Do not stretch, outline, shadow, bevel, or gradient-fill the mark.
- Prefer typographic identity over ungoverned logo proliferation.

SECTION 5

Token architecture

5.1 Why tokens are the source of truth

The DTCG 2025.10 format is a strong fit for 561 because it explicitly expresses platform-agnostic design decisions in JSON and is intended for exchange among tools. The Group should not invent a private color/token serialization unless a missing requirement can be demonstrated.

NORMATIVE

Core visual primitives **MUST** be represented in `tokens/561.tokens.json` or a successor DTCG-compatible source. Medium renderers **MUST** derive from tokens wherever the DTCG model can represent the decision without loss.

5.2 Four token tiers

Tier	Example	Rule
Primitive	color.ink	Physical value with minimal semantics.
Semantic	text.primary, evidence.witness	Meaning, not medium. Prefer aliases to primitives.
Habitat	habitat.coesite. surface	Mission-specific expression inheriting semantic roles.
Medium	latex.rule.strong, web.gutter	Rendering decisions needed because physical media differ.

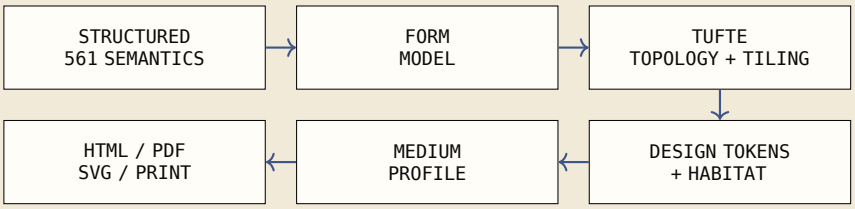
5.3 Naming rules

Token names **SHOULD** read from broad category to specific role. Do not encode implementation accidents such as `blue-3-button-hover` when the stable semantic role is `action.focus`. Names should survive a palette redesign.

GOOD / BAD

Prefer	Avoid
color.witness	`medium-blue`
`evidence.rule`	`1px-gray-line`
`type.evidence`	`jetbrains`
`space.7`	`big-padding`

5.4 Projection contract



TUFTE remains the Group presentation/projection engine boundary. The acronym expansion is currently disputed in the registry, so this Bible treats TUFTE as the canonical name and does not settle the expansion by fiat.

SECTION 6

Color system

CORE PALETTE

Few devices. Scarce salience. Strong semantic roles.

6.1 Core colors

Ink
#0A0A0A

Ink soft
#494742

Bone
#F2EAD8

Rule
#CDC6B7

Witness
#3D5588

Dark ground
#120C02

Alarm
#FF4530

Paper white
#FFFDF7

6.2 Semantic roles

- **Ink**: primary text, primary rules, institutional marks, strong borders.
- **Bone**: default Group ground and warm paper field.
- **Witness**: citations, links, verification, selected state, evidence paths, informational emphasis.
- **Alarm**: action, exception, consequence, hazard, or a deliberately singular high-salience moment.
- **Ink soft**: metadata and secondary apparatus that remains readable.

- **Rule**: low-salience dividers, table rules, scaffolding.

NORMATIVE

Witness and alarm colors are semantic. They **MUST NOT** be distributed merely to make a page "more colorful." A local composition **SHOULD** have one primary accent role; if both witness and alarm are active, their meanings must be unambiguous.

6.3 Computed contrast

The following table is generated from the token source by the R build step. Normal-text PASS uses a 4.5:1 threshold; large/UI PASS uses 3:1 as a practical minimum. The table is not a substitute for full WCAG conformance, but it catches the easiest palette mistakes automatically.

Foreground	Background	Ratio	Normal text	Large/UI
ink	bone	16.53:1	PASS	PASS
witness	bone	6.15:1	PASS	PASS
alarm	bone	2.85:1	FAIL	FAIL
alarm	ink	5.79:1	PASS	PASS
ink	paper-white	19.46:1	PASS	PASS
bone	dark-ground	16.25:1	PASS	PASS
witness	paper-white	7.24:1	PASS	PASS

EXCEPTION / CONSEQUENCE

Alarm red on bone fails text contrast. Therefore **alarm red is not a body-text color on the default bone ground**. Use it as a rule, mark, fill with ink text, or place it against ink where contrast is sufficient. This is exactly why design computation belongs in the build rather than in taste alone.

6.4 Dark mode

Dark mode is a real medium profile, not inversion. The recovered static apex uses a brown-black ground rather than neutral black. A conforming dark profile should preserve the semantic relationships among ink, witness, alarm, rule, and surface while recomputing contrast, not mechanically swap foreground/background values.

SECTION 7

Typography

7.1 Three channels

The 561 type system is intentionally triadic.

Channel	Preferred family	Job
Display	Space Grotesk	Institutional force, hero marks, chapter openers, large wayfinding, short declarations.
Reading	Roboto Slab	Long-form prose, explanations, essays, reports, durable reading surfaces.
Evidence	JetBrains Mono	Folios, metadata, labels, code, measurements, identifiers, receipts, compact navigation.

This PDF uses locally available fallbacks when the preferred faces are not installed. A production repository should either self-host properly licensed font files or install them in the controlled build environment; renderers must never silently fetch web fonts at runtime.

7.2 Specimen

The institution speaks in mass.

The reading face carries sustained argument without turning every paragraph into a poster. Its slab structure keeps enough mechanical character to belong beside diagrams, tables, and operating evidence.

```
locus=urn:561:visual:typography / state=candidate / measure=112mm / build=deterministic
```

7.3 Scale

Use a constrained modular scale rather than arbitrary font sizes. The candidate base is 10.25pt for A4 print and 1.0625rem for long-form web reading, with a 1.25 ratio for major steps. Extreme display sizes may escape the scale when used as singular typographic architecture rather than paragraph hierarchy.

Role	Nominal print size	Use
Evidence micro	7–8pt	folio, metadata, provenance, figure apparatus
Caption / note	8.5–9pt	captions, callouts, compact tables
Body	10.25pt	long-form text
Lead	12–14pt	opening summaries, pull statements
Section	18–20pt	second-level hierarchy
Chapter	30–34pt	major section identity
Hero	60pt+	singular marks / title plates only

NORMATIVE

Never use font weight, uppercase, letterspacing, color, and size simultaneously unless the element genuinely carries multiple hierarchical roles. Evidence labels are the main exception: small uppercase mono with tracking is an intentional apparatus convention.

7.4 Line length and rhythm

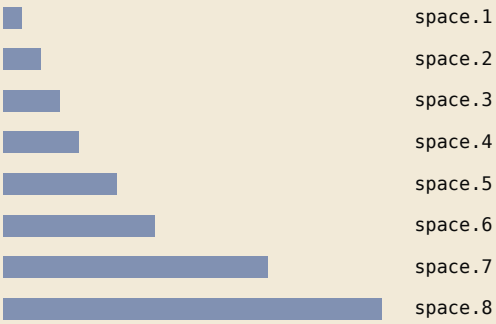
Long-form web text targets approximately 42rem as recovered from the current theme. Print targets approximately 112mm for the primary reading measure on A4. Narrower measures are acceptable for notes and compact components; wider measures require columns or a different reading task.

SECTION 8

Spatial system and grid

8.1 Spacing ladder

The spacing system is a named ladder, not a bag of numbers. Medium profiles map abstract steps into physical values. On web, step 7 corresponds to the recovered 56px default gutter; on small screens the current implementation reduces the gutter to 24px.



8.2 Grid law

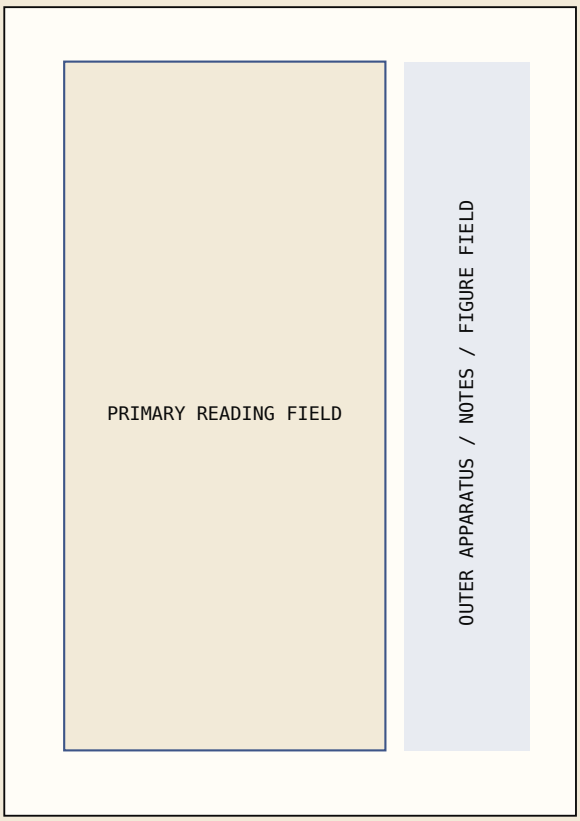
The grid is a geometric relationship among content, not a visible cage. It should explain alignment across titles, prose, figures, tables, annotations, and controls.

NORMATIVE

A 561 composition **MUST** use a small number of repeatable alignment lines. It **SHOULD** expose generous negative space around major hierarchy changes and compress spacing inside strongly related evidence clusters.

8.3 A4 print profile

This Bible uses ISO A4 as the canonical standards-manual trim: 210 by 297mm. The candidate profile uses 22mm inner, 38mm outer, 20mm top, and 24mm bottom margins. The wider outer field is reserved for apparatus, notes, and visual breathing room. US Letter is a required alternate production target, but the system does not treat page-size changes as simple global scaling.



A4 conceptual anatomy (not to scale)

8.4 Web profile

Web content can expand to a 1200px institutional frame, but reading text remains measure-constrained. Wide space is used for navigation, side apparatus, visualizations, or topology—not to stretch prose across the viewport.

SECTION 9

Rules, edges, shape, and texture

9.1 Rules are structural language

Recovered 561 work repeatedly uses borders and rules to establish hierarchy. This is preferable to nested shadows, cards, and soft floating surfaces.

Rule	Candidate weight	Meaning
Hairline	0.3pt / 1px-ish	low-salience alignment, table interior
Standard	0.8pt	component edge, section divider
Strong	2pt	major block / active structure
Institutional	8px web equivalent	folio/colophon/identity bar, rare

9.2 Corners

Core 561 geometry is square-cornered. Radius is zero in the current theme specification. Small rounding may be admitted within a habitat where it expresses a material or interaction need, but indiscriminate rounded-card UI is not house style.

9.3 Texture

The current web theme uses a subtle 4px radial dot texture on the bone background. This is admissible because it gives the digital surface a physical grain without competing with content. Print defaults to flat ground to prevent unnecessary toner/ink and moiré.

NORMATIVE

Texture **MUST** be low-contrast and subordinate. It **MUST NOT** reduce text contrast or become the primary signifier of “futurism.”

SECTION 10

Information design

10.1 Tuftean, not Tufte cosplay

The 561 system uses “Tuftean” to mean an information ethic: make evidence legible; maximize meaningful comparison; integrate words, numbers, and graphics; avoid decoration that obscures reasoning. It does not mean copying the visual appearance of a particular Tufte book.

10.2 Captions are evidence

A figure caption should answer at least two of these questions: What am I looking at? What does it demonstrate? Where did it come from? When was it observed? What is uncertain? What should I compare?

CAPTION ANATOMY

Figure 7. Deployment witness topology. The diagram distinguishes semantic source, renderer, deployment actuator, and public readback so a live URL cannot silently become identity authority. Source: 561 web-estate consolidation registry, observed 2026-10-06.

10.3 Tables

Tables are for comparison, not boxed decoration. Prefer light horizontal rules, aligned numeric columns, explicit units, and a meaningful row order. Avoid alternating zebra stripes unless rows are genuinely hard to track.

10.4 Figures and diagrams

Preferred figure types include:

- topology and dependency diagrams;
- annotated screenshots when interface evidence is needed;
- scientific/engineering schematics;

- small multiples and comparative matrices;
- maps when geography is causal or operational;
- timelines when sequence matters;
- deterministic Bauhaus/geometric image macros when a concept benefits from abstraction.

EXCEPTION / CONSEQUENCE

Decorative generative imagery should never be the default solution to an empty space. Exact typography, UI, diagrams, data graphics, logos/marks, and provenance-sensitive assets belong to deterministic tools or evidence-bearing media.

10.5 Data visualization

A chart **MUST** carry units, source/provenance, and a readable encoding without depending solely on hue. Direct labels are preferred to legends when practical. The core palette is too small for many-category charts; data visualization therefore uses a separate functional sequence derived for perceptual separation, while reserving witness/alarm semantics for their system meanings.

SECTION 11

Imagery and visual assets

11.1 Asset order of operations

The Group's visual-asset service should operate as a demand-led internal service, not an image-generation hobby. The preferred order is:

1. identify consuming office/project and downstream use;
2. check for an existing admissible asset;
3. rank the missing asset by reuse, communication value, revenue/operational value, urgency, and uniqueness;
4. use the least lossy production method: deterministic diagram/vector, documentary image, fabricated object/render, or generative image where actually appropriate;
5. record provenance, rights, accessibility text, and correction/supersession state.

11.2 Photography

Photography should feel documentary, situated, and material. Prefer real environments, people doing real work, instruments, field conditions, ecological scenes, manufacturing/operations, civic places, and evidence of systems in use. Avoid generic stock-photo affect.

11.3 Generated imagery

Generated imagery may be appropriate for abstract editorial concepts, exploratory world-building, or reusable visual macros. It is not appropriate for evidence, exact UI, exact product representation, marks/logos, or contexts whose provenance must be documentary.

NORMATIVE

Every externally used generated image **SHOULD** have a provenance sidecar and alt text. A model-generated asset **MUST NOT** be presented as documentary evidence of a real event, place, person, product, or measurement.

11.4 Deterministic visual macros

The most 561-native future path is a library of deterministic SVG/TikZ macros: geometric compositions, topology plates, process diagrams, texture fields, isometric evidence diagrams, and editorial abstractions derived from semantic parameters. This makes visual language inspectable, diffable, and reusable across HTML and LaTeX.

SECTION 12

Interaction and motion

12.1 Motion has a job

Historical 561 web work used page entrance and link pulse motion, with reduced-motion handling. The candidate rule is stricter: motion is admitted when it orients, reveals causality, confirms state, or helps the user preserve context.

NORMATIVE

Ambient motion **MUST NOT** be required to perceive structure. Every nonessential animation **MUST** be disabled or substantially reduced when the user requests reduced motion.

12.2 Focus is visible architecture

Keyboard focus should look intentional rather than browser-error accidental. The current apex uses a 3px outline with offset. WCAG 2.2's focus guidance reinforces the principle that focus must remain visible and sufficiently contrasted.

12.3 Links

Witness blue is the normal interactive/evidence color where color use is appropriate, but links must also have non-color affordance in long-form reading contexts. Hover decoration may add motion; focus must remain obvious without hover.

SECTION 13

Accessibility and legibility

13.1 Accessibility is upstream

Accessibility is not a compliance layer applied after art direction. The visual system must make accessible choices the easiest choices.

NORMATIVE

Renderer profiles **MUST** define: minimum contrast rules, visible focus treatment, reduced-motion behavior, minimum readable text, logical heading hierarchy, link affordance, table semantics, alt-text/provenance hooks, print behavior, and zoom/reflow resilience appropriate to the medium.

13.2 Color

The R build computes a basic contrast table for critical core pairs. This must be extended into a linter that rejects forbidden foreground/background token combinations.

13.3 Type and measure

Readable body type, adequate leading, and constrained line length are mandatory. Extreme tracking, all-caps, or microtext are reserved for short apparatus labels.

13.4 PDF accessibility

The long-term target is tagged, navigable PDF with meaningful reading order, bookmarks, alt text for figures, embedded fonts, and archival conformance where required. PDF/A addresses long-term visual preservation; PDF/UA should be evaluated for accessibility conformance once the LaTeX toolchain can produce it reliably. The first priority is a stable, inspectable build rather than falsely claiming conformance we have not validated.

SECTION 14

Component grammar

14.1 Four primary containers

This Bible demonstrates four semantic containers:

NORMATIVE

Normative blocks carry requirements. They are black-ruled because the primary distinction is authority, not emotion.

EVIDENCE / RATIONALE

Evidence blocks carry provenance, rationale, measurements, or observations. Witness blue indicates that the reader is looking at support for a claim.

EXCEPTION / CONSEQUENCE

Exception blocks carry hazards, contradictions, failures, or consequences. Alarm red is a marker at the edge, while the reading text remains ink-on-paper for contrast.

EXAMPLE

Examples are pedagogical and non-normative unless explicitly stated. They are deliberately quieter than requirements.

14.2 Cards are not a default

Use a bordered card only when an object has an independent identity, state, and interaction boundary. Do not turn ordinary paragraphs, feature lists, or navigation into card grids simply because a component framework makes that easy.

14.3 Forms

Forms should resemble serious instruments: clear label, instruction, control, unit/-format where applicable, validation, and consequence. Avoid placeholder-as-label. Error color must be paired with text/icon/state, never color alone.

14.4 Status and metadata

Status labels use the evidence channel and compact rectangular framing. They should describe state with unambiguous words: `CANDIDATE`, `VERIFIED`, `DEPRECATED`, `BLOCKED`. Avoid vague badges such as “new” when a date or state is available.

SECTION 15

Math, code, proof, and technical documents

15.1 Technical material is first-class

561 publications routinely contain equations, code, schemas, measurements, standards references, proofs, diagrams, and execution evidence. The design system must make these feel native rather than pasted in.

15.2 Mathematics

Mathematics should remain typographically conventional enough to read. Do not force house display typography into formulas. Surrounding labels, theorem state, proof status, and provenance can use the house evidence channel.

MATHEMATICAL PLATE

Let X be a measured state space and $T : X \to X$ an admitted transformation. A visual witness is not merely a picture of T; it is a projection $P(T, x)$ whose provenance binds the displayed marks to the source state x and the transformation version.

$$\text{semantic state} \xrightarrow{T} \text{form model} \xrightarrow{P} \text{marks}$$

15.3 Code

Listing 15.1. A document using the 561 LaTeX profile.

```
\documentclass{report}
\usepackage{561design}
\begin{document}
\chapter{Measured work}
\begin{evidencebox}
This paragraph is evidence-bearing apparatus.
\end{evidencebox}
\end{document}
```

15.4 Proof/status apparatus

Research documents should distinguish conjecture, observation, derivation, proof, replication, supersession, and editorial commentary. Visual styling may make these states easier to scan, but it must never upgrade epistemic status. A beautiful box cannot turn a conjecture into a theorem.

SECTION 16

Habitat profile: 561 Group apex

HABITAT / WHOLE INSTITUTION

The public institutional house.

The apex should feel like the whole system: corporate/institutional, federative, mathematically literate, and durable. It is the calmest habitat because it must host the widest range of topics.

NORMATIVE

Apex defaults: bone ground; ink primary; witness blue evidence; alarm red exception/action; display/reading/evidence triad; square geometry; hard folio bars; large typographic “561” mark permitted; generous hierarchy; minimal ornament.

16.1 Material cues

Archival standards manual, field dossier, civic institution, systems map, laboratory report, technical catalog. The apex can borrow cues from every subordinate habitat but should not impersonate any one of them.

16.2 Web chrome

Recovered web chrome—folio bar, giant 561 hero, monospaced descriptor, inked colophon—is a strong candidate for the apex. The Bible recommends preserving the idea while moving implementation into TUFTE rather than binding identity to one static HTML package.

SECTION 17

Habitat profile: Lenticule Science

HABITAT / SCIENCE

Clinical brightness, curiosity, instruments, impossible clarity.

Lenticule Science may riff on the institutional-science feeling associated with fictional mega-laboratories, but it must be original: no copied marks, props, type treatments, or trade dress. The useful qualities are bright controlled space, instrument labeling, experiment state, test chambers, optical surfaces, and a little unnerving confidence in measurement.

17.1 Expression

- Bias toward paper white and bone with witness blue, black instrument labels, and restrained glass/gray fields.
- Use grids, calibration marks, specimen identifiers, plots, equations, lab-note marginalia, and experiment state.
- Let curiosity and evidence carry the weirdness; avoid generic neon “science” graphics.
- Diagrams and data are the dominant image class.

SCIENCE HEADER

LENTICULE SCIENCE / EXPERIMENTAL RECORD 26-104

Reality rewards curiosity.

apparatus: optical / state: replicated / witness: three independent runs

SECTION 18

Habitat profile: Red Cup Engineering

HABITAT / ENGINEERING

Concrete, steel, instrumentation, test stands, consequence.

Red Cup should feel like serious applied engineering inside a slightly impossible research facility. The reference is not game fandom; it is the material logic of industrial research: hazard boundaries, maintenance stencils, equipment IDs, cutaways, test receipts, and systems that can fail in concrete ways.

18.1 Expression

- Ink, bone, concrete gray, and alarm red are natural dominant materials.
- Use witness blue mainly for verification, documentation, and successful readback rather than as decorative tech-blue.
- Prefer section numbers, equipment labels, cutaway diagrams, interface contracts, test matrices, and failure envelopes.
- Strong rules and visible assembly logic are encouraged.

ENGINEERING RULE

Red is allowed to be more physically present in Red Cup than at the Group apex, but it still means consequence: hazard, actuator, stop, exception, or the “cup” identity moment. Do not make every engineering surface red.

SECTION 19

Habitat profile: Coesite Operations

HABITAT / OPERATIONS

Subterranean granite, limestone, containment, logistical calm.

Coesite should feel like monumental infrastructure below grade: granite/limestone mass, long corridors, durable signage, inventories, access layers, incident boards, maintenance states, and institutional strangeness. It may brush against backrooms/SCP-like spatial uncanniness, but horror is not the default.

19.1 Expression

- Dark ground, bone type, low-saturation rule colors, and hard illuminated status marks.
- Floor/zone identifiers, location codes, manifests, maintenance stamps, inventories, route maps, and environmental readings.
- Large empty stone-like fields can carry scale, but information-bearing surfaces should remain crisp.
- Avoid distressed “secret bunker” cosplay; operations must look usable.

COESITE / ZONE 04 / MATERIAL HANDLING

ALIGN. ENABLE. DELIVER.

SECTION 20

Habitat profile: Causeway Public Services

HABITAT / CIVICS

Impeccable public-service design. Calm authority, clear obligations.

Causeway should be the least self-conscious visual system: extremely legible, trustworthy, accessible, procedural, and humane. It can be beautiful, but the beauty comes from service clarity, good typography, excellent forms, and visible accountability.

20.1 Expression

- Paper white/bone, ink, witness blue, generous whitespace, stable navigation.
- Clear next actions, eligibility, deadlines, costs, ownership, appeal/correction paths, and contact methods.
- Public information should resemble the best civic systems, not consumer SaaS marketing.
- Forms and notices get first-class design attention.

CIVIC SERVICE PANEL

CAUSEWAY PUBLIC SERVICES

People. Places. Together.

What this service does. One sentence in plain language.
What you need. Inputs, eligibility, and time.
What happens next. Explicit consequence and route for correction.

SECTION 21

Habitat profile: Koios.News

HABITAT / INDEPENDENT MEDIA

News identity is related to the house, but has stricter local invariants.

Koios is a sibling public enterprise and should not be swallowed by the apex identity. Its current local visual direction uses royal “008 blue” (#000088) for social-image identity, deterministic isometric/30-degree construction, and a five-blade aperture in which the southeast blade is fully blue. Faux grain is not part of the current direction.

NORMATIVE

Koios social imagery **MUST** use its local royal blue rather than substituting 561 alarm red. Article-page deterministic steganographic imagery remains a separate system and should not be replaced by generic social-image art.

This is a useful model for Group governance: local habitats can have strict visual invariants while inheriting common provenance, accessibility, and build discipline.

SECTION 22

Habitat profile: Turntide Islands

HABITAT / CIVIC CLIMATE CULTURE

Provisional profile: inherit the house until an evidence-backed local system exists.

Turntide needs recognizable public identity for advocacy, membership, governance, and supporter participation, but this Bible does not invent a full palette or logo in the absence of settled evidence. Its provisional profile therefore inherits the 561 civic/-house grammar and adds only mission-level composition guidance.

22.1 Provisional expression

- Civic, participatory, climate-aware, human-scale; no apocalypse porn.
- Use maps, island/shoreline/topology diagrams, governance flows, membership pathways, public invitations, and practical climate adaptation artifacts.
- Supporter/membership materials should be welcoming and legible even to people who do not want governance participation.
- Once Turntide has enough real use, derive its local tokens from recurring artifacts rather than designing an ornamental subbrand first.

SECTION 23

LaTeX as a first-class renderer

23.1 Why LaTeX

LaTeX is unusually well matched to 561's needs: mathematical typesetting, bibliographies, cross-references, long technical documents, deterministic builds, stable pagination, and a programmable style layer. It can also act as a target for a more general TUFTE form model rather than as the authoring source for every semantic object.

23.2 Repository layout

Listing 23.1. Candidate repository layout.

```
561-visual-design-bible/
  .github/workflows/build-pdf.yml
  R/derive_tokens.R
  tokens/561.tokens.json
  generated/                  # R-derived build products
  561design.sty               # LaTeX renderer profile
  design-bible.tex            # this specification/specimen
  README.md
  Makefile
  LICENSE                      # choose/admit explicitly
```

23.3 Package responsibilities

`561design.sty` owns page geometry, font channels/fallbacks, color bindings, hierarchy, folios, tables, semantic callouts, list rhythm, links/bookmarks, code styling, and deterministic visual primitives. It should not own semantic content or business-specific prose.

23.4 Renderer API

A small stable API is preferable to exposing raw package internals:

- `specbox` — normative requirement;
- `evidencebox` — observation/provenance/rationale;
- `alarmbox` — hazard/exception/consequence;
- `examplebox` — non-normative illustration;
- `\FSOLabel{}` — compact evidence label;
- `\FSOStatus{}` — explicit state badge;
- `\FSOSectionPlate{}{}{}` — deterministic section plate;
- `\FSOColorChip{}{}{}` — palette specimen.

NORMATIVE

Document authors **SHOULD** use semantic macros rather than direct colors, font switches, or arbitrary TikZ styling. Direct styling is allowed only inside renderer implementation or a documented habitat extension.

SECTION 24

R as design computation

24.1 Why R belongs in the build

R is not being used merely because it can draw charts. It gives the design system a compute stage between raw tokens/data and final publication. That stage can test, derive, summarize, and generate visual evidence before LaTeX renders it.

24.2 Current build computation

The included `R/derive_tokens.R` script:

1. parses the DTCG token JSON;
2. validates that required core colors exist and carry valid hexadecimal extensions;
3. computes WCAG relative-luminance contrast ratios for critical pairs;
4. emits LaTeX `\definecolor` declarations and specification macros;
5. emits a contrast conformance table consumed by this PDF;
6. emits a deterministic validation receipt with no wall-clock timestamp.

Listing 24.1. Contrast math inside the build.

```
channel_linear <- function(c)
  ifelse(c <= 0.04045, c / 12.92, ((c + 0.055) / 1.055)^2.4)

contrast <- function(a, b) {
  l <- sort(c(luminance(a), luminance(b)), decreasing = TRUE)
  (l[[1]] + 0.05) / (l[[2]] + 0.05)
}
```

24.3 Next computations to add

- forbidden token-pair linting and habitat override validation;
- CIE/OKLCH palette derivation for web dark/light profiles;
- line-length and type-scale reports;

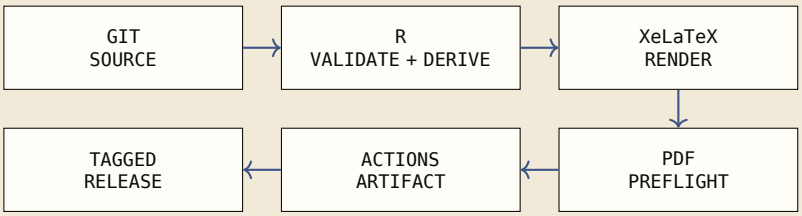
- chart-theme generation for base R/ggplot2 and LaTeX/PGFPlots;
- SVG/TikZ macro generation from semantic/topological data;
- visual regression contact sheets;
- PDF metadata and conformance receipts;
- automated comparison between CSS and LaTeX token outputs.

SECTION 25

GitHub Actions: source to PDF artifact

25.1 Build contract

The repository includes an Actions workflow that builds on changes to the Bible, tokens, LaTeX renderer, R compute, or workflow itself. It also supports manual dispatch and tagged releases.



25.2 CI acceptance

A CI build should eventually fail when:

- token validation fails;
- a required contrast pairing violates its threshold;
- XeLaTeX emits a fatal error;
- the PDF has no pages or is corrupt;
- required fonts are not embedded;
- visual golden pages differ unexpectedly;
- accessibility/archival profile checks regress where those profiles are claimed.

NORMATIVE

The PDF is an artifact, not the source. Releases may attach PDFs, but the accepted source remains tokens + semantic content + renderer + compute scripts + build workflow.

SECTION 26

Conformance profiles

26.1 Profile matrix

Profile	Scope	Minimum requirements
V-CORE	any 561 visual	semantic colors, type channels, no parallel identity, provenance rules, accessibility intent
V-DOC	LaTeX/PDF	A4/Letter geometry profiles, hierarchy, figures/-captions, tables, PDF metadata, embedded fonts
V-WEB	HTML/TUFTE	responsive gutters/measure, dark/light profile, focus, reduced motion, semantic markup
V-SCI	Lenticule	V-CORE + lab/instrument/evidence habitat rules
V-ENG	Red Cup	V-CORE + engineering/hazard/instrumentation habitat rules
V-OPS	Coesite	V-CORE + operations/zone/inventory habitat rules
V-CIVIC	Causeway	V-CORE + service clarity/forms/public trust rules
V-MEDIA	Koios etc.	V-CORE provenance/accessibility + admitted local identity invariants

26.2 Claiming conformance

A produced artifact may state a profile and version, e.g. `V-DOC / V-CORE / 0.4-candidate`. It should not simply say "561 compliant" once multiple profiles exist.

26.3 Golden specimens

The conformance repository should maintain a small suite of golden outputs:

- one-page letter/report;
- long-form technical paper;
- data table + chart + figure plate;
- public-service form/notice;

- science experiment record;
- engineering test report;
- operations status/incident sheet;
- web article and dashboard-like operator surface.

These are not templates to copy blindly; they are regression witnesses.

SECTION 27

Governance and versioning

27.1 Design changes are engineering changes

The Bible should be versioned like infrastructure. A palette or type change can affect thousands of artifacts and accessibility behavior; it deserves an explicit change record, migration notes, and tests.

27.2 Decision classes

Class	Example	Required handling
Patch	fix incorrect hex, typo, broken macro	tests + patch release
Minor	new component or habitat token	rationale + examples + backwards-compatible renderer update
Major	replace type system or semantic palette roles	migration plan + cross-medium review + visual regression baseline reset
Experimental	new image macro or interaction pattern	isolated namespace; cannot redefine core tokens

27.3 No silent drift

Web, PDF, decks, social images, and software UI will drift unless a machine checks them. The Group should prefer transformations from shared tokens over copy/paste. When a medium needs a nonportable value, the exception belongs in that medium's profile, not in an undocumented local stylesheet.

27.4 Visual debt

Visual debt includes duplicate token values, untracked local palettes, inaccessible color combinations, ad-hoc page geometry, typography bypasses, stale logo files, ambiguous asset rights, missing alt text, inconsistent correction state, and templates whose implementation no longer matches the Bible.

SECTION 28

Migration plan from what exists now

28.1 Phase 1: admit the Bible repository

1. Place this source package under Group-controlled Git history.
2. Choose the permanent repository boundary: dedicated visual-system repository or an admitted section of the Group superrepository.
3. Run R + XeLaTeX + preflight on every relevant change.
4. Publish build artifacts and checksums from CI.

28.2 Phase 2: unify tokens

Translate recovered web palette/type/layout seeds into the DTCG source. Then update the web theme/TUFTE renderer to consume generated outputs rather than owning its own manually typed color values.

28.3 Phase 3: cross-medium proof

Take one semantic document and render it from the same form/topology source into:

- HTML/TUFTE;
- LaTeX/PDF;
- deterministic SVG figure/plate;
- plain/accessible fallback representation.

Compare hierarchy and semantic roles rather than pixel equality.

28.4 Phase 4: habitat packages

Add Lenticule, Red Cup, Coesite, Causeway, Koios, and other local token groups only after each has real artifacts to test. Local systems extend the house; they do not fork it.

28.5 Phase 5: conformance service

Build a visual linter that agents can call before publishing. It should report exact failures: token drift, contrast, font embedding, missing provenance, invalid profile claim, unapproved logo use, PDF preflight, and golden-image regression.

SECTION 29

Implementation checklist

29.1 For a new LaTeX document

1. Use XeLaTeX and `561design.sty`.
2. Run `R/derive_tokens.R` before compilation.
3. Do not define ad-hoc brand colors in the document source.
4. Use semantic callouts rather than direct tcolorbox styling.
5. Keep long-form measure within the document profile.
6. Caption every figure with evidentiary purpose and source/provenance.
7. Use mono apparatus for IDs/state/measurements, not for entire prose sections.
8. Preflight the PDF, render representative pages, and inspect typography/overflow.
9. Embed fonts and record the profile/version in metadata or colophon.

29.2 For a new visual component

1. State its semantic purpose.
2. Check whether an existing component already covers the purpose.
3. Define states, content constraints, accessibility behavior, and failure mode.
4. Use semantic tokens.
5. Add one normal, one stressed, and one failure specimen.
6. Add a conformance test or golden output.
7. Only then add it to the renderer API.

29.3 For a new habitat

1. Name the mission and material world, not a trendy aesthetic.

2. Identify what must remain inherited from V-CORE.
3. Collect real artifacts and recurring needs.
4. Add the smallest local token extension that explains them.
5. Test across at least web + document outputs before declaring it stable.

SECTION 30

Token registry

Token	Value	Meaning
color.ink	#0A0A0A	Primary text/mark/rule.
color.bone	#F2EAD8	Primary warm ground.
color.witness	#3D5588	Evidence, reference, verification, links/selection.
color.alarm	#FF4530	Exception, action, consequence; not normal text on bone.
color.ink-soft	#494742	Secondary apparatus.
color.rule	#CDC6B7	Low-salience structure.
color.dark-ground	#120C02	Dark-mode / deep-operations ground.
color.paper-white	#FFFDF7	High-key figure/component ground.
type.family.display	Space Grotesk	Primary institutional display channel.
type.family.reading	Roboto Slab	Long-form reading channel.
type.family.evidence	JetBrains Mono	Labels, code, IDs, receipts, measurements.
type.measure.reading	42rem	Web long-form target.
type.measure.print-body	112mm	A4 long-form target.
layout.web.max-width	1200px	Institutional web frame.
layout.web.gutter	56px	Default large-screen web gutter.
layout.web.mobile-gutter	24px	Current narrow-screen gutter.
shape.radius	0px	Core square geometry.
motion.standard-ms	300ms	Ordinary interaction transition ceiling/target.
motion.entrance-ms	500ms	Existing historical entrance timing reference; use only when orientation benefits.

SECTION 31

Build and release recipe

31.1 Local build

```
Rscript R/derive_tokens.R .
latexmk -xelatex -interaction=nonstopmode -halt-on-error design-bible.tex
```

31.2 Expected products

```
design-bible.pdf
build/ or generated/ derivation receipts
CI artifact containing PDF + source checksum
(optional) tagged release attachment
```

31.3 Preflight

A release pipeline should run, at minimum:

```
qpdf --check design-bible.pdf
pdfinfo design-bible.pdf
pdffonts design-bible.pdf
pdftotext design-bible.pdf - >/dev/null
```

Then render representative pages to images and compare to accepted golden images. A future archival job may add veraPDF once a specific PDF/A profile is formally claimed.

SECTION 32

References and precedents

Bibliography

[1] NASA. NASA Brand Center: Graphic Standards. https://www.nasa.gov/nasa-brand-center/brand-guidelines/

[2] U.S. Web Design System. Design principles. https://designsystem.digital.gov/design-principles/

[3] U.S. Web Design System. Design tokens. https://designsystem.digital.gov/design-tokens/

[4] GOV.UK Design System. Styles. https://design-system.service.gov.uk/styles/

[5] GOV.UK Design System. Spacing. https://design-system.service.gov.uk/styles/spacing/

[6] GOV.UK Design System. Type scale. https://design-system.service.gov.uk/styles/type-scale/

[7] Design Tokens Community Group. Design Tokens Format Module 2025.10. https://www.w3.org/community/reports/design-tokens/CG-FINAL-format-20251028/

[8] W3C Web Accessibility Initiative. Web Content Accessibility Guidelines 2.2. https://www.w3.org/TR/WCAG22/

[9] IBM Carbon Design System. 2x Grid. https://www.carbondesignsystem.com/building-blocks/foundations/2x-grid/guidelines

[10] ISO. ISO 216:2007 — Writing paper and certain classes of printed matter. https://www.iso.org/standard/36631.html

[11] PDF Association. ISO 19005 / PDF-A. https://pdfa.org/resource/iso-19005-pdfa/

[12] 561 Group. Web estate control-root README. https://github.com/561-group/561-group/blob/main/sites/README.md

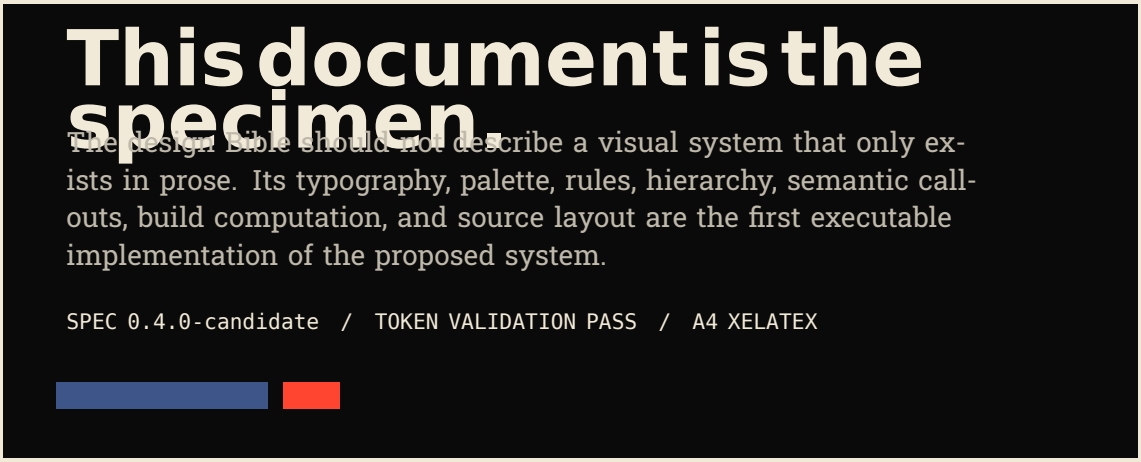
[13] 561 Group. Web estate registry. https://github.com/561-group/561-group/blob/main/sites/registry.json

[14] 561 Group. 561 Group website theme implementation. https://github.com/561-group/561-group-website-theme/blob/main/src/index.mts

[15] emsenn. Historical 561 Hugo theme stylesheet. https://github.com/emsenn/561-hugo-theme/blob/main/static/style.css

[16] Red Cup Engineering. Web TUFTE GUI Services. https://github.com/red-cup-engineering/web-tufte-gui-services-section

Colophon



The next institutional move is not to add more prose. It is to admit this source under Group Git history, wire the web/TUFTE renderer to the same token source, render a matched HTML/PDF specimen, and let real use pressure the candidate rules into a stable 1.0.